

Design of a Sliding Mode Control for a DC-to-DC Buck-Boost Converter

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Abstract—This paper studies DC-to-DC buck-boost converter, as well as of the most important features associated with the sliding mode control, such as adding robustness to the system with respect to variations of its parameters and external disturbances applied to this converter.

Index Terms—buck-boost converter, sliding mode control

I. INTRODUCTION

THE control of switched power converters, dc-dc, is a very active area of research, both in power electronics and automatic control theory. The importance of this research is that many industrial applications currently require characteristics such as speed of response, stability against disturbances and high efficiency.

A dc-dc converter is a device that converts an unregulated voltage input to a regulated voltage output by a switching action.

The buck-boost converter is considered a variable structure system that consists of two continuous subsystems, nonlinear, modeled by a finite number of first order differential equations coupled together. On the other hand, the magnitude of the voltage output may be higher or lower compared with the magnitude of the voltage input and of opposite polarity, depending on the arrangement of electronic components in the circuit, as shown in Figure. Furthermore, the voltage output can vary linearly from 0 to $(+/-)\infty$ (for an ideal converter) adjusting the duty cycle of semiconductor, such that if it is 50%, the voltage output is equal to the voltage input in magnitude.

The theory of sliding mode control applied to systems of variable structure was introduced in the 50's in the former Soviet Union by Emelyanov and other partners, it has been since developed in nonlinear systems, multi input-output systems discrete, stochastic systems, etc. Switching is performed by logic which is a function of system states. The control action resulting from this switching law is a discontinuous function of the states. A particular operation mode is obtained when the switching occurs at very high frequency, for power electronics in the order of KHz, restricting the trajectory of the system states to a variety in the state space. This operating mode is called mode or sliding regime, due to the sliding surface which must meet the necessary and sufficient condition of sliding mode and show its invariance against disturbances.

II. THE BUCK-BOOST CONVERTER DC-TO-DC

Steady-state analysis allow us to know the behavior of power electronic components and obtain the formulas for sizing of the converter in function of their waveforms. Furthermore, the control of system is represented by nonlinear state space model of the form $\dot{x} = f(x) + g(x) \cdot u$ which will serve the purposes of control.

A. Permanent State Operation and Continuous Mode

If the current through the inductor L never falls to zero during a commutation cycle, the converter is said to operate in continuous mode. But in some cases, the amount of energy required by the load is small enough to be transferred in a smaller time than the whole commutation period, in this case, the current through the inductor falls to zero during part of the period, so the converter is said to operate in discontinuous mode.

Because of the switching transistor in the non-inverted configuration of buck-boost converter, Figure 1 (b), is distinguished as a multivariable system that switching between two circuits which make the coil to work in continuous mode. During activation cycle of the transistors, as shown in Figure 2, there is a current flowing through the inductor which stores energy. In this cycle, the transistors (MOS1 and MOS2) enter into conduction, due to pulse at the gate, this is a high value of control signal, at the same instant the two diodes (D1 and D2) do not lead because they are reverse biased so the charging cycle of the coil ends.

During the off cycle of the transistors, Figure 3, the current flow in the inductor is interrupted, by a low pulse from the control signal that reaches the gate MOS1 and MOS2, which

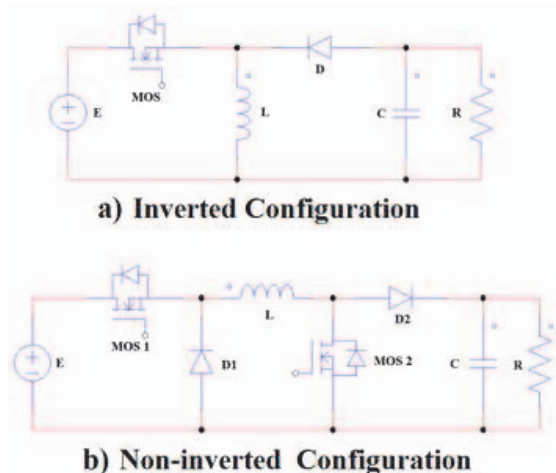


Figure 1: Buck-Boost Converter dc-to-dc, asynchronous

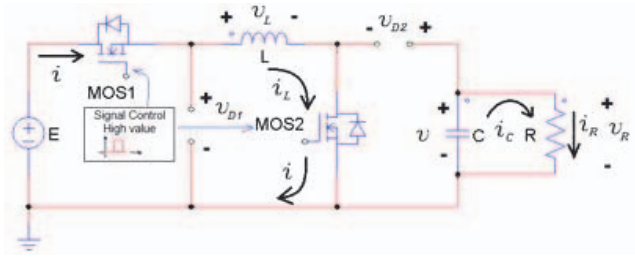


Figure 2: Activation cycle of the transistors

produces the simultaneous deactivation of the transistors. Furthermore, due to the effect of the electromotive force against the coil, the direction of the current flow discharges in the inductor, is the same as in the load circuit, however the polarity of the voltage at its terminals is reversed, allowing energy to be delivered to the load by simultaneous activation of the diodes which are biased directly.

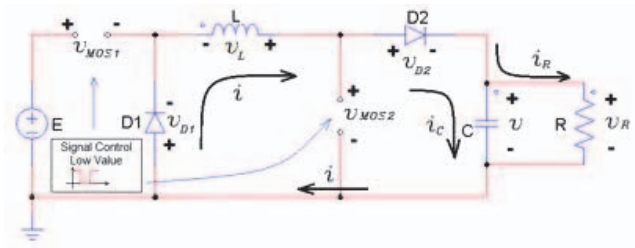


Figure 3: Off cycle of the transistors

This behavior provides the waveforms that are shown in Figure 4. Based on these waveforms, the equations describing the average current per switching cycle applicable to this converter are obtained as:

$$I_{DS,prom} = \frac{I_1 + I_2}{2} \cdot D \quad (II.1)$$

$$I_{D,prom} = \frac{I_1 + I_2}{2} \cdot (1 - D) \quad (II.2)$$

$$I_{L,prom} = \frac{I_1 + I_2}{2} \quad (II.3)$$

Considering $I_{DS} = I_E$, $I_D = I_O$ (due to ideal behavior of the capacitor; open circuit) and $T_{on} = D \cdot T$ (T is the period of the switching), then holds:

$$I_O = \frac{(1 - D)}{D} \cdot I_E \quad (II.4)$$

Where:

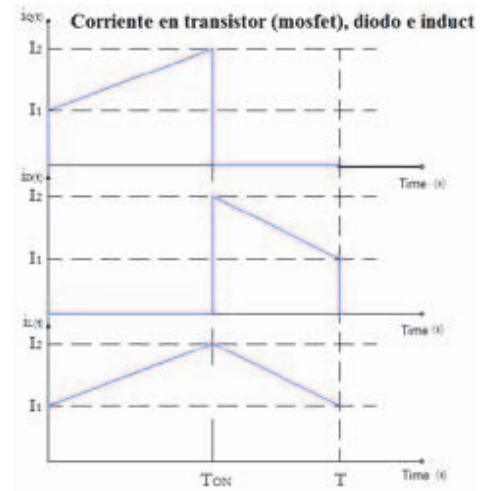
D is the duty cycle of the switching signal.

I_O is the output current or load current.

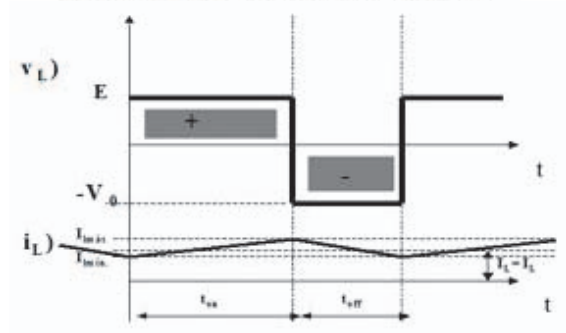
I_E is the input current provided by the source.

On the other hand, taking into account losses, depreciable, for a circuit where the input power equals the output power, then holds:

$$E \cdot I_E = V_O \cdot I_O \quad (II.5)$$



a) Current of transistors (mosfet), diode and inductor.



b) Relación Tensión - Corriente en el inductor

Figure 4: Waveforms resulting from the operation of the converter.

The value of the output voltage is V_O , the voltage drop at the terminals of the load resistance. Substituting equation II.4 in equation II.5 and solving for V_O , the following relationship is obtained:

$$V_O = \frac{D}{1 - D} \cdot E \quad (II.6)$$

B. Design of the Converter

1) *Inductor and Capacitor Case:* The most critical part in the design of a converter is the coil as it is difficult, in practice, getting an inductor with a value as high as possible to reduce the ripple component of peak current, according to equation II.7 and then negotiate the value of capacitor, such that it has a low ripple voltage and simultaneously does not come into resonance with the inductor. Furthermore, the coil must be constructed to support the high frequency (kHz) in power electronics. Finally, coil and capacitor must meet:

$$\Delta I = \frac{E \cdot D}{f \cdot L} \quad (II.7)$$

$$\Delta V_C = \frac{I_O \cdot D}{f \cdot C} \quad (II.8)$$

For the critical case in which the converter works on the border between modes of continuous and discontinuous

conduction, the minimum inductance value is:

$$L_{min} = \frac{E.D.(1-D)}{2.I_{O,min-prom}.f} \quad (II.9)$$

2) *Semiconductor selection case:* To select both the transistor and the diodes is important to first consider the voltage which is observed when the transistors are open, and in the case of the current, is the maximum observed in the winding of the inductor, taking into account that the frequency of operation is high, it show be taken as the peak current reference and not the average, because the most significant losses are caused by switching and not due to conduction, then it must meet:

$$\begin{cases} V_{DS,max,mos1} > E + V_{D1} \\ V_{DS,max,mos2} > V_L - V_{D1} \\ I_{DS} > I_{L,max} \end{cases} \quad (II.10)$$

For the case of the diodes will be:

$$\begin{aligned} I_{D1} &= I_{D2} > I_{L,max} \\ V_{RRM,D1} &> E \\ V_{RRM,D1} &> V_O \end{aligned}$$

It is necessary to notice that it is difficult to achieve low ripple, mainly due to the output and input currents are discontinuous. On the other hand, a high value inductor (mH) that supports the high frequency current, is difficult to achieve. The switch also needs to be able to drive a high peak current and must supports a large voltage block, commercially available devices such as MOSFET transistor meet this requirement.

C. Modeling of the Converter

For the case of modeling the non-inverted configuration of Figure 1 (b) is considered because it is desired for the implementation. A system of variable structure as in the case of the converter having two states, for when the switching signal u takes a high value equal to one and the other is different when u takes on a value under equal to zero. Figure 5 (b). If $u = 1$, the equations are:

Voltage: $E = L \cdot \frac{d}{dt}i$, where the variable i is the current in the inductor and solving for the derivative of the current is:

$$\frac{d}{dt}i = \frac{E}{L} \quad (II.11)$$

Current: $-C \frac{d}{dt}v = \frac{v}{R}$, where the variable v is the voltage on the capacitor, which means that the capacitor is discharged into the resistor, and similar to the above clears, in this case the derived voltage:

$$\frac{d}{dt}v = -\frac{v}{R.C} \quad (II.12)$$

Figure 5(c), if $u = 0$, the equations are:

In voltage: $L \cdot \frac{d}{dt}i + v = 0$, and solving:

$$\frac{d}{dt}i = -\frac{v}{L} \quad (II.13)$$

In current: $i = C \cdot \frac{d}{dt}v + \frac{v}{R}$, and solving:

$$\frac{d}{dt}v = \frac{i}{C} - \frac{v}{R.C} \quad (II.14)$$

Taking into account the two possible values of u and equations II.11, II.12, II.13 and II.14, then we has the following system:

$$\frac{d}{dt}i = -(1-u) \cdot \frac{v}{L} + u \cdot \frac{E}{L} \quad (II.15)$$

$$\frac{d}{dt}v = (1-u) \cdot \frac{i}{C} - \frac{v}{R.C} \quad (II.16)$$

As shown in Figure 5 (a)

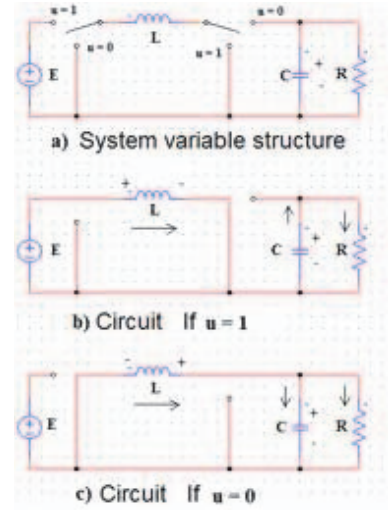


Figure 5: Converter operation for mathematical modeling

III. SLIDING MODE APPLIED TO THE BUCK-BOOST CONVERTER DC-DC

The sliding mode is sufficiently described in the literature, in this article we only mentions the specific application Buck-Boost converter dc-dc. To start with the analysis of the system, equations II.15 and II.16 must respond to the form:

$$\dot{x} = f(x) + g(x) \cdot u \quad (III.1)$$

or

$$\dot{x} = A \cdot x + \alpha + u \cdot (B \cdot x + \beta) \quad (III.2)$$

Replacing: $x_1 = i$ and $x_2 = v$, then holds: $\frac{d}{dt}i = \dot{x}_1$ y $\frac{d}{dt}v = \dot{x}_2$. Therefore the system of equations II.15 and II.16 can be represented by:

$$\begin{cases} \dot{x}_1 = -(1-u) \cdot \frac{x_2}{L} + u \cdot \frac{E}{L} \\ \dot{x}_2 = (1-u) \cdot \frac{x_1}{C} - \frac{x_2}{R.C} \end{cases} \quad (III.3)$$

and in matrix form it can be represented as:

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{bmatrix} 0 & -\frac{1}{L} \\ \frac{1}{C} & -\frac{1}{R.C} \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{bmatrix} u \cdot \frac{E}{L} \\ 0 \end{bmatrix} \quad (III.4)$$

As seen, equation III.4 and III.3 correspond to the system forms III.2 and III.1 respectively.

The autonomous nonlinear system described by the equation III.1 corresponds to III.5 and satisfies that, $x \in X$, with $X \subset \mathbb{R}^n$ open set, this is the states vector; $u : \mathbb{R}^n \rightarrow \mathbb{R}$ is the control action; and f and g local vector fields are sufficiently smooth and defined in X , with $g(x) \neq 0, \forall x \in X$. It define the switching function h as a smooth function $h : X \rightarrow \mathbb{R}$, with dh gradient different from zero in X , also called sliding surface, which should be proposed for validation, for this particular case it is proposed:

$$h(x) = x_1 - x_{1d} \quad (\text{III.5})$$

where: x_{1d} is a constant that can choose the operator as set point.

A control law variable structure can be obtained by forcing the control action u to take a value of two possible, depending on the sign of the switching function $h(x)$:

$$u = \begin{cases} u^+(x) & \text{si } h(x) \geq 0 \\ u^-(x) & \text{si } h(x) < 0 \end{cases} \quad u^+(x) \neq u^-(x), \quad (\text{III.6})$$

with $u^+(x)$ and $u^-(x)$ smooth functions X and, without loss of generality, one can accept that satisfy $u^+(x) > u^-(x)$ ($u^+(x) = 1, u^-(x) = 0$). It is said that there is a sliding regime on the sliding surface when as a result of the control law III.6 the system reaches the sliding surface and held locally in its environment. Furthermore, it must meet controlled vectorial fields $(f + g \cdot u^+)$ and $(f + g \cdot u^-)$ pointing locally to the sliding surface, (Figure 6) [3]:

$$\begin{cases} \lim_{h \rightarrow 0^+} \langle dh, f + g \cdot u^+ \rangle = \lim_{h \rightarrow 0^+} L_{f+g \cdot u^+} h < 0 \\ \lim_{h \rightarrow 0^-} \langle dh, f + g \cdot u^- \rangle = \lim_{h \rightarrow 0^-} L_{f+g \cdot u^-} h > 0 \end{cases} \quad (\text{III.7})$$

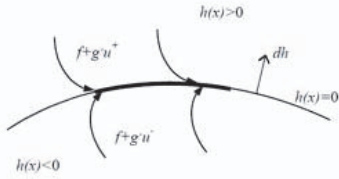


Figure 6: Sliding mode on a sliding surface

Therefore it calculates:

$$\begin{aligned} L_f h &= \frac{\partial h}{\partial x_1} \cdot f_1 + \frac{\partial h}{\partial x_2} \cdot f_2 = \langle \nabla h, Ax + \alpha \rangle \\ &= \left\langle \begin{bmatrix} 1 & 0 \end{bmatrix}, \begin{bmatrix} -\frac{x_2}{C} \\ \frac{x_2}{RC} \end{bmatrix} \right\rangle = -\frac{x_2}{L} \end{aligned} \quad (\text{III.8})$$

$$L_g h = \left\langle \begin{bmatrix} 1 & 0 \end{bmatrix}, \begin{bmatrix} \frac{x_2}{L} + \frac{E}{L} \\ -\frac{x_1}{C} \end{bmatrix} \right\rangle = \frac{x_2}{L} + \frac{E}{L} \quad (\text{III.9})$$

This applies to the equivalent control method to solve the problem created by the discontinuity on the right side of the differential equation (III.1)[1], it is based on recognizing that $\dot{h}(x) = 0$ as a necessary condition for confining the path of

states about the variety $h(x) = 0$, this invariance condition is described in III.10. It is defined, then the equivalent control action $u_{eq}(x) = 0$ as the law of soft feedback control (fictitious) for which the sliding surface is an invariant local system range III.3 or III.4, as seen in Figure 7.

$$\begin{cases} h(x) = 0 \\ \dot{h}(x) = \langle \nabla h, f + g \cdot u_{eq}(x) \rangle = L_{f+g \cdot u_{eq}} h = 0 \end{cases} \quad (\text{III.10})$$

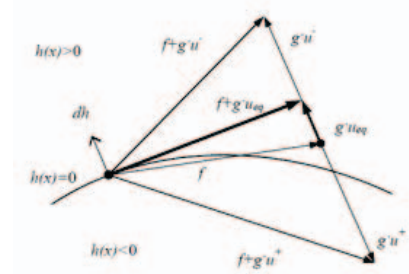


Figure 7: Equivalent Control Method

Then the equivalent control of III.10 is obtained directly evaluated in $h = 0$, so that by replacing III.8 and III.9 we have:

$$u_{eq} = -\frac{L_f h}{L_g h} = -\frac{-\frac{x_2}{L}}{\frac{x_2}{L} + \frac{E}{L}} = \frac{x_2}{x_2 + E} \quad (\text{III.11})$$

The conditions of existence of sliding mode, transversality condition, equation III.12 and the necessary and sufficient condition without loss of generality, equation III.13:

$$\langle dh, g \rangle = L_g h \neq 0 \quad (\text{III.12})$$

$$\min(u^-(x), u^+(x)) < u_{eq}(x) < \max(u^-(x), u^+(x)) \quad (\text{III.13})$$

The stability criterion used is the Lyapunov direct method because it allows to analyze the entire system. For this, consider the scalar quantity defined by equation III.14 is consider as semipositive (greater than or equal to zero), which represents a certain "energy" instantaneous error with respect to the variety S , is identically zero on the variety S . [2].

$$\varepsilon(x) = \frac{1}{2} \cdot h^2(x) \geq 0 \quad (\text{III.14})$$

A recommended strategy to achieve the desirable condition $h(x) = 0$ from a neighborhood of S , is to run control actions, $u \in \{0, 1\}$ resulting in a strict decrease of the function $\varepsilon(x)$ respect of time t . This is achieved by influencing the system so that the rate of change of $\varepsilon(x)$ with respect to time is strictly negative. [2]. Then,

$$\frac{d}{dt}(\varepsilon(x)) = \frac{1}{2} \cdot \frac{d}{dt}(h^2(x)) = h(x) \cdot \dot{h}(x) < 0 \quad (\text{III.15})$$

Considering $\dot{h}(x) = L_{f+g \cdot u} h(x) = L_f h(x) + L_g h(x) \cdot u$.

Then,

$$\begin{aligned} h(x) \cdot \dot{h}(x) &= h(x) \cdot [L_f h(x) + L_g h(x) \cdot u] \\ &= h(x) \cdot [L_f h + L_g h \cdot u - \\ &\quad - L_g h \cdot u_{eq} + L_g h \cdot u_{eq}] \\ &= h(x) \cdot [0 + L_g h \cdot (u - u_{eq})] \end{aligned}$$

Therefore,

$$h(x) \cdot \dot{h}(x) = h(x) \cdot [L_g h \cdot (u - u_{eq})] < 0 \quad (\text{III.16})$$

Replacing III.6, III.9 and III.11 in III.16, then:

If $u = 1$, $h(x) < 0$, then $\dot{h}(x) = -h(x)$

$$\begin{aligned} h(x) \cdot \dot{h}(x) &= -h(x) \cdot \left[\left(\frac{x_2 + E}{L} \right) \cdot \left(1 - \frac{x_2}{x_2 + E} \right) \right] \\ h(x) \cdot \dot{h}(x) &= -h(x) \cdot \left[\frac{E}{L} \right] < 0 \end{aligned}$$

where E is the supply voltage and L is the inductance, which are positive

If $u = 0$, $h(x) > 0$, then $\dot{h}(x) = h(x)$

$$\begin{aligned} h(x) \cdot \dot{h}(x) &= h(x) \cdot \left[\left(\frac{x_2 + E}{L} \right) \cdot \left(0 - \frac{x_2}{x_2 + E} \right) \right] \\ h(x) \cdot \dot{h}(x) &= h(x) \cdot \left[-\frac{x_2}{L} \right] < 0 \end{aligned}$$

where x_2 is the output voltage and L is the inductance. Therefore, due to the foregoing demonstrates the stability of III.3 on the sliding surface III.5.

IV. SIMULATION

The program used to simulate the Buck-Boost converter dc modeled by system III.3, was Scilab 5.

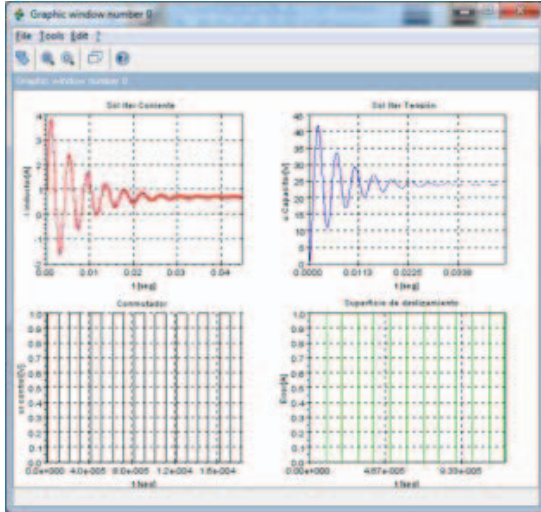


Figure 8: Simulation results of Buck-Boost DC Converter

To solve the differential equation it was used the Runge-Kutta numerical method of fourth order. Furthermore, using inductor, $L=4.40[\text{mH}]$; power supply, $E=24[\text{V}]$; resistance load, $R=70[\Omega]$; capacitor, $C=100[\mu\text{F}]$; and, frequency $50[\text{KHz}]$ with 50% duty cycle 50%. The results shown in figure 8 are achieved.

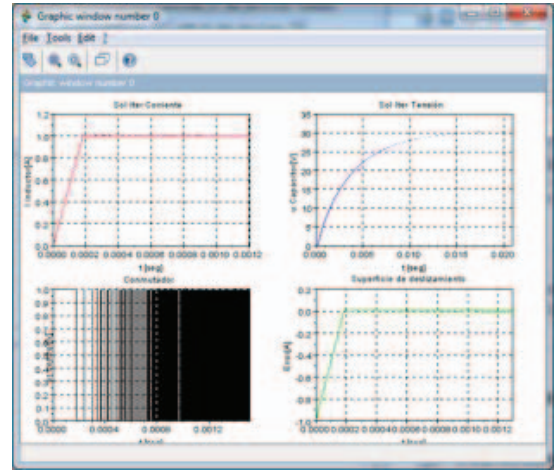


Figure 9: Simulation results of sliding mode control

By implementing the control in sliding regime in the same program with a current value in the desired coil $x_{1,d} = 1[\text{A}]$, the results shown in Figure 9 are obtained.

V. IMPLEMENTATION

A. Buck-Boost Converter DC-DC

For the prototype implementation of the converter input data from the simulation was taken the electronic components which were previously selected by the design considerations outlined above in section II.2, so that the components are:

Inductor: A ferrite toroid was selected as shown in Figure 10 (a), due to the work frequency is superior to 1 KHz, and the form was chosen to reduced parasitic effects as capacitances. The winding was made with enamelled wire # 18AWG up to 4 [A] in coil as shown in Figure 10 (b). The inductance value is 4.4 [mH].

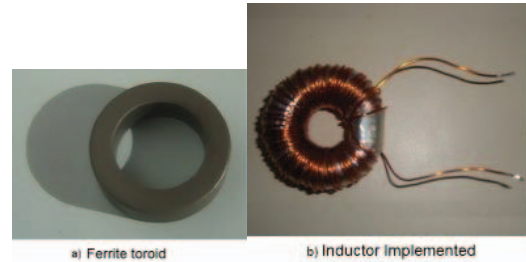


Figure 10: Inductor

Setup: Figure 11 shows the proposed elements used in the simulation, considering the capacitor to 100 [V]. Furthermore IRF840 mosfet transistors with their coolers, fast diodes FR307 and as a load resistor (rheostat) whose characteristics are 70 [Ω] and 3 [A].

B. Sliding Mode Control

Figure 12 shows on the left, the circuit that allows activation of mosfet transistors, which contains an integrated circuit IR2130, one floodgate 7404 and one optocoupler 6N137. On the right side, the circuit that generates the control contains

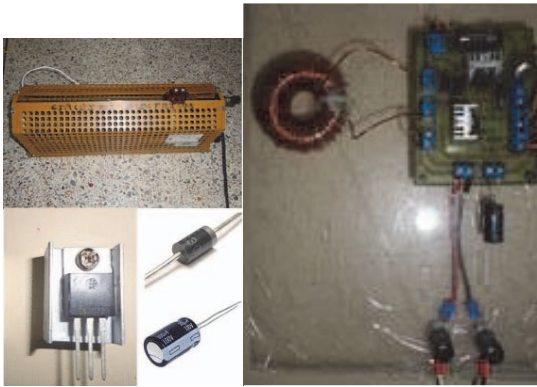


Figure 11: Buck-Boost Converter DC and its components.

a current sensor ACS712T that are found on board that it has coupling and gain necessary for getting the signal which will enters by conversion channel analogous-digital, to the microcontroller PIC18F2550, which has implemented the sliding surface, through C code.

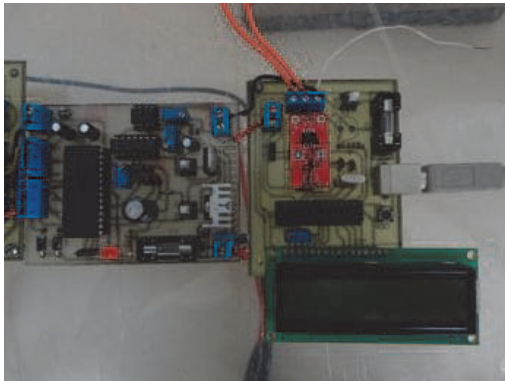


Figure 12: Controller setup

C. Final Setup

Figure 13 shows the entire assembly circuit, the buck-boost converter DC controlled by sliding mode which will serve to do tests.

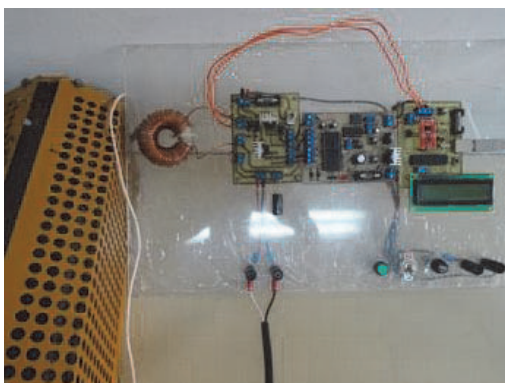


Figure 13: Full mount

VI. CONCLUSIONS

- The converter output voltage is a function of the duty cycle variation of signal switches, transistors, and only of that value, so that for $D < 0.5$ behaves as a reducer and for $D > 0.5$ behaves as elevator.
- The implementation of the Buck-Boost DC converter is cheap and its components can be easily obtained in the market except for ferrite toroid.
- A disadvantage of this converter is that the requirements of the components are greater than those needed for another type of converter, for example, the boost converter.
- One benefit of sliding mode control is that the designer do not need the converter transfer function to start the analysis, the differential equation describing the behavior of the converter and responds to the form $\dot{x} = f(x) + g(x) \cdot u$ can be used as starting point.
- A set of sliding surfaces can be applied, however, not all meet the conditions of existence of the sliding mode, analysis of stability and robustness, so that not always the first surface proposed is ok. Therefore there is not a technique for selecting that surface.
- The sliding mode control is continuous, as seen from the perspective of switching this is possible at infinite frequency, for this case and according to the simulation would be around three hundred kilohertz (300 [kHz]). The disadvantage is found in power electronic devices without this limitations. Therefore, the control will work at finite frequency, leading to the existence of a hysteresis band witch is located around the sliding surface producing a ripple effect on the values of state variables, current in the inductor and capacitor voltage, for the converter case implemented this effect is minimal and therefore not considered significantly.
- Another advantage of the sliding mode control is robustness against disturbances that may be located in the loading and / or the power supply.
- In implementing the power circuit contains an antiparallel diode to the transistor mosfet, also known as a diode free-wheel that allows free movement to reduce the switching losses that occur when the transistor operates at high frequency such as for the implementation on this project.

ACKNOWLEDGEMENTS

The authors would like to thank Dr.C. Alexander Fernandez Correa, professor of the Polytechnic University CUJAE, who with his knowledge greatly facilitated the implementation of this work.

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